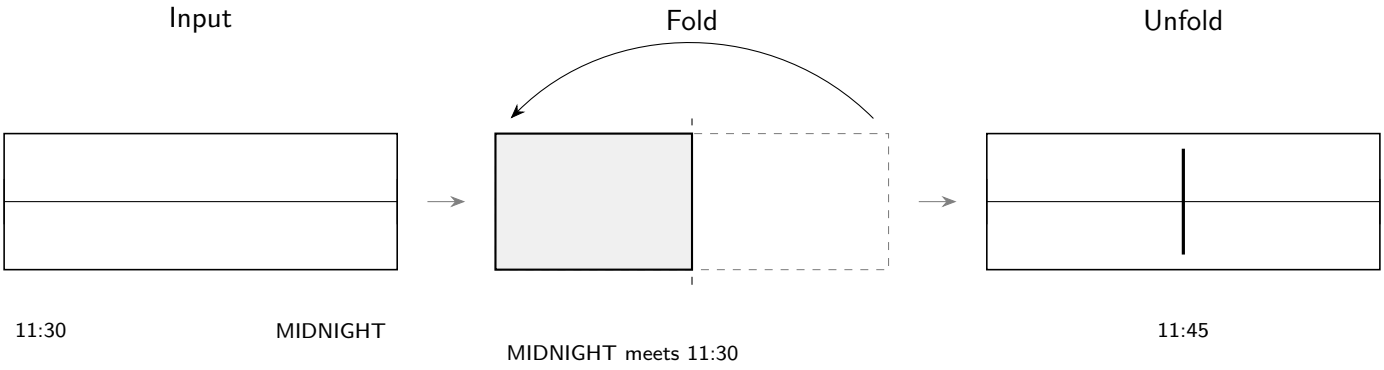
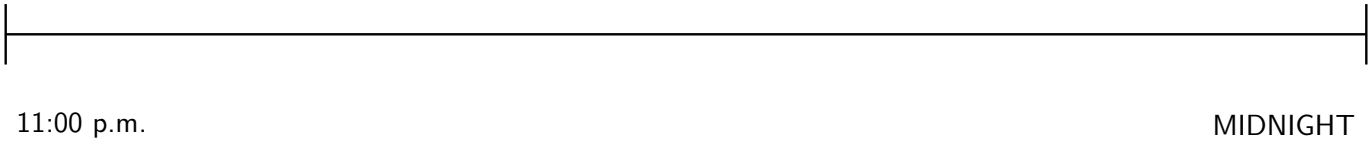
A light is OFF at 11:00 p.m. At 11:30 it switches ON; at 11:45 it switches OFF. It keeps switching halfway from the previous switching time to midnight, forever before midnight.

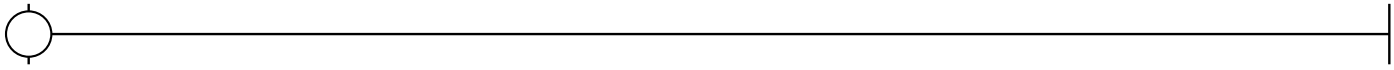
Use the ON/OFF card for the light. Mark switching times on a paper strip from 11:00 p.m. to MIDNIGHT. Flip the card at every switching mark.



Problem 1: Make the first four switching marks on a paper strip. Play the rule with the card. Can the rule have a last switching mark before midnight?



Problem 2: Use the same switching marks. The card starts OFF. A counter starts at START. At each switching mark, flip the card and move the counter halfway from its present place toward END. As the rule keeps going, does the card settle on one side? Does the counter get closer and closer to one place?

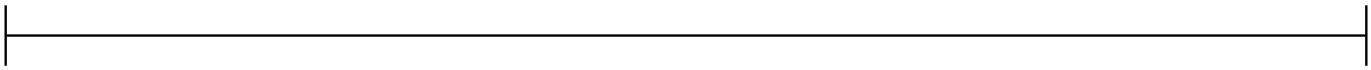


START END

Card

Counter

Problem 3: Use the same switching marks. Make up a flipping rule for a second card that does settle on one side. Make up a moving rule for the counter that does not get closer and closer to one place.

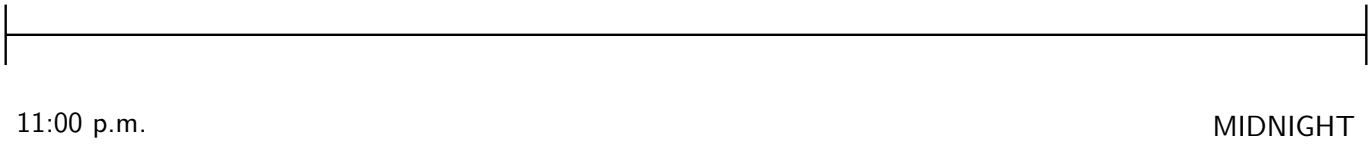


START END

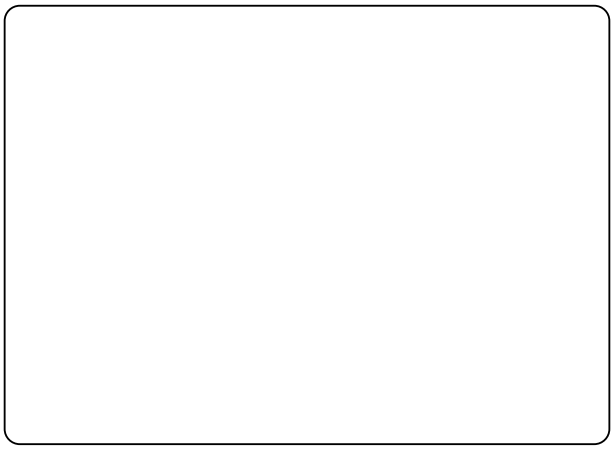
Card rule

Counter rule

Problem 4: What, if anything, does the switching rule alone tell you to show at midnight?



MIDNIGHT

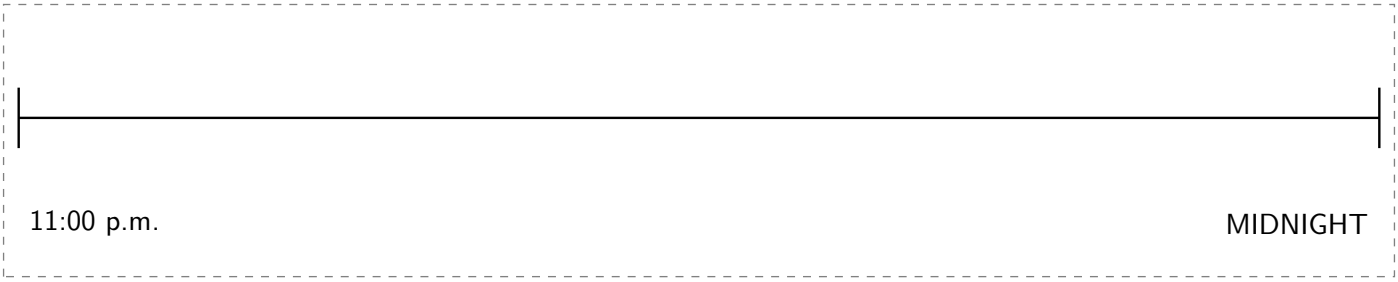


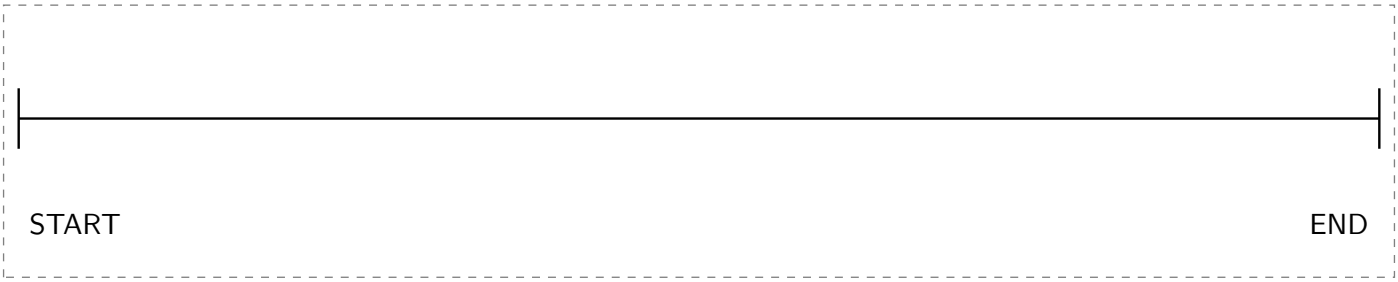
Problem 5: The switches before midnight stay the same. Make two different extra rules for the card at midnight. Decide whether either extra rule follows from the original switching rule.

At midnight:

At midnight:

Problem 6: Someone adds this rule: "At midnight, use the state that the earlier states approach." Try that rule for the ON/OFF card and for the moving counter. Does it assign a state in each case?





ON

OFF